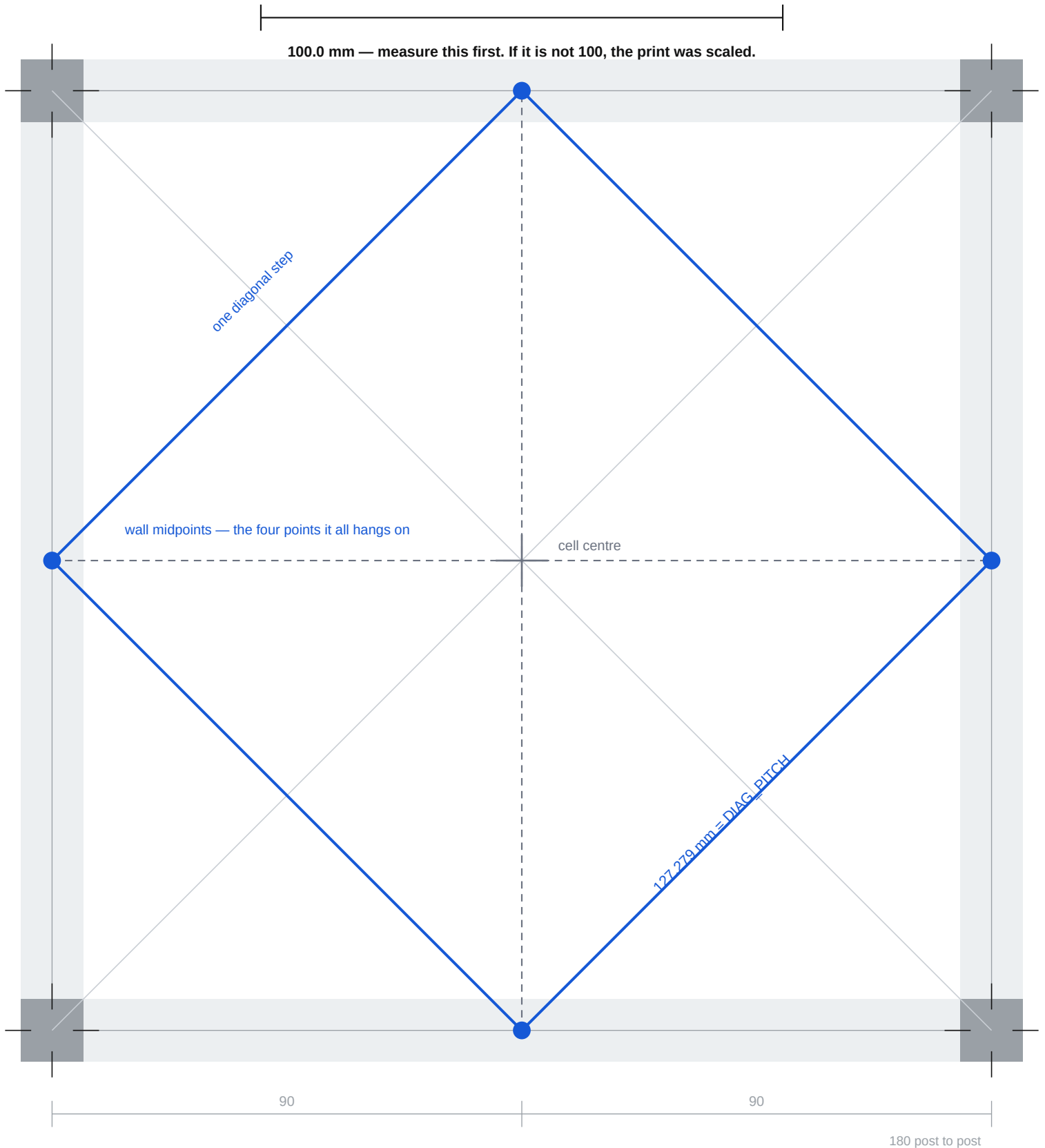


E4 — cell template, 1:1

lay it in a cell, square it to the posts, mark through the four midpoints



- diagonal running line**
joins the four wall midpoints. The lane she is on between an SD45 and a DS45.
- orthogonal centrelines**
the straight lanes — 90 mm from the boundary, 84 mm clear of a wall face.
- corner to corner**
post to post. For checking squareness and 45 by eye — she never drives on it.

To mark it out: find each wall midpoint (90 mm from either post) and join the four. That diamond passes a post on every side without touching one, which is the whole point of it. The cell centre is NOT on it — that is what the 45s have to bridge.

E4 — turn tuning

one page per row. Every number below is derived from the row itself — nothing here was measured.

row	name	ω	R	tangent	+½ cell	needs	has	out by	page
1	SS90ER	170	101	101	—	101	100	-1	3
0	SS90EL	170	101	101	—	101	100	-1	4
3	SS90R	170	101	101	—	101	100	-1	5
2	SS90L	170	101	101	—	101	100	-1	6
7	SD45R	95	181	75	90	165	120	-45	7
6	SD45L	95	181	75	90	165	120	-45	8
8	DS45L	95	181	75	—	75	120	+45	9
9	DS45R	95	181	75	—	75	120	+45	10
14	DD90L	273	63	63	—	63	63	agrees	11
15	DD90R	273	63	63	—	63	63	agrees	12
5	SS180R	191	90	—	—	—	90	+0	13
4	SS180L	191	90	—	—	—	90	+0	14

Four of the twelve rows already agree with themselves. The 45s do not, and they cannot both be right while they share one number: an SD45 is booked in at a cell centre while its two lanes cross at the wall midpoint half a cell earlier, so it wants 165, and a DS45 turns at its own lattice point and wants 75. Both hold 120 today.

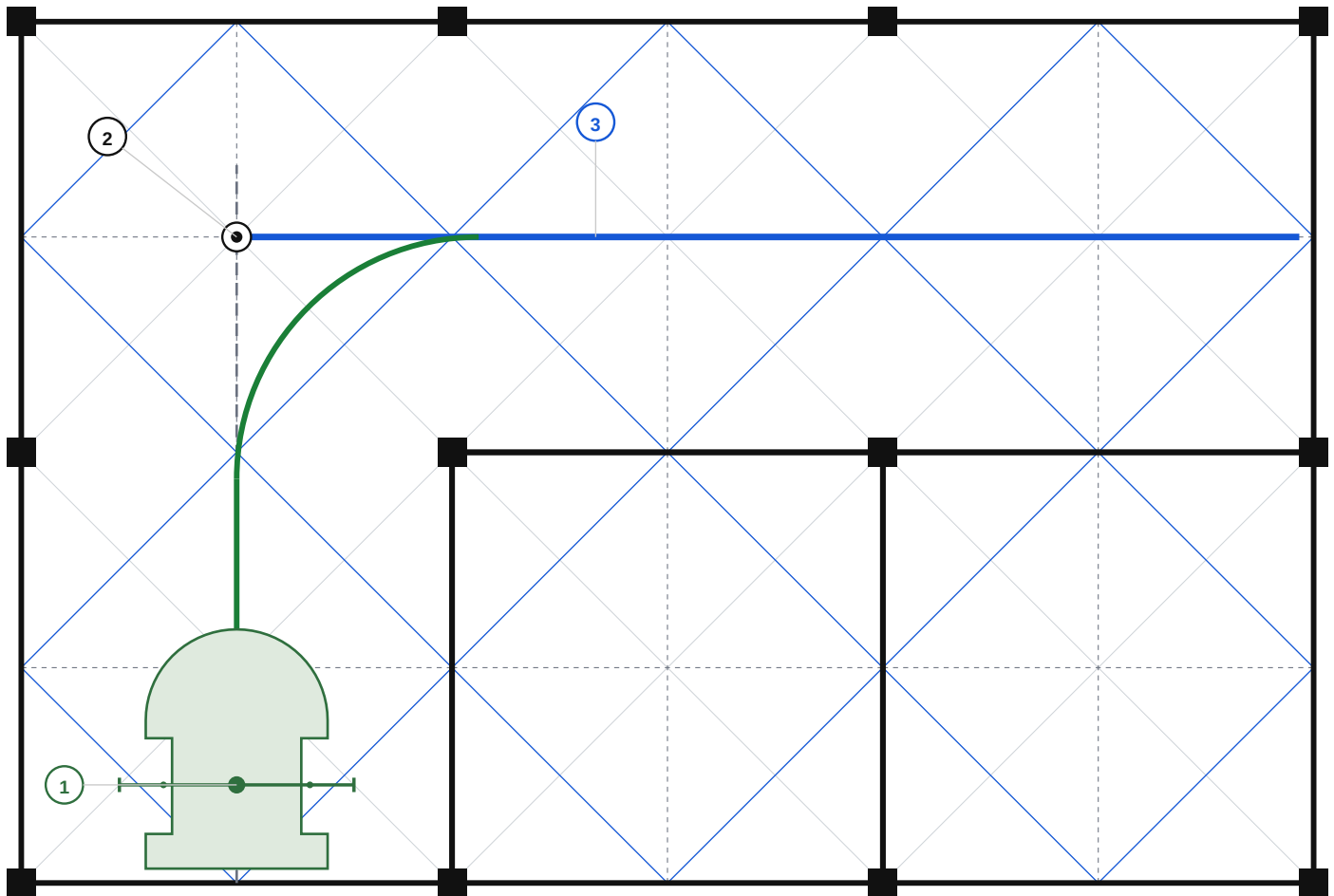
A running order that does not waste an afternoon: gyro first, then the 90s (rows 0–3) because they are the only rows that have ever met a floor and they are the check on everything else. Then SD45 and DS45 separately — never chained — because a lateral error out of the first moves the start of the second. Then DD90, which has 127 mm either side of it instead of 180 and is the tightest thing she does. The 180 last: its omega is forced by the cell pitch, so there is nothing to search for.

Trim the OFFSET, not omega. Omega sets the radius, and the radius has to fit the space it is turning in; the offset is free. Both move her, at the rates each page gives.

Each page carries the exact three lines to type into Turn tuning, with the corrected offset already in them. S saves the table to EEPROM; nothing is permanent until it does.

Row 1 — SS90ER, a 90 out of a straight

the search 90 — she stops and senses either side of it, so it need not chain



the lane — measure to this the turn it should drive

SEL,1 POS,1 ARC,300,-90,170,2500,101,90 — the table holds 100; this line corrects it

commanded speed	300 mm/s	
omega	170 deg/s	the only thing that sets the radius
$R = v / \omega$	101 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(45^\circ)$	101 mm	either side of where the lanes cross
+ half cell	0 mm	two straight lanes cross AT the cell centre, so nothing is added
entry_offset it needs	101 mm	so the arc is tangent to both lanes
entry_offset it has	100 mm	out by -1 mm
lead-in it drives now	129 mm	out of the start cell, then the arc
lead-in once corrected	128 mm	she starts turning 1 mm earlier

What to read off the floor

1 Back her against the wall, facing north. The marked line across her is the AXLE LINE — her centre of rotation, and the point every distance on this page is measured from.

2 The two lanes cross AT the cell centre for a 90, so there is no half cell to add here — the tangent is the whole offset.

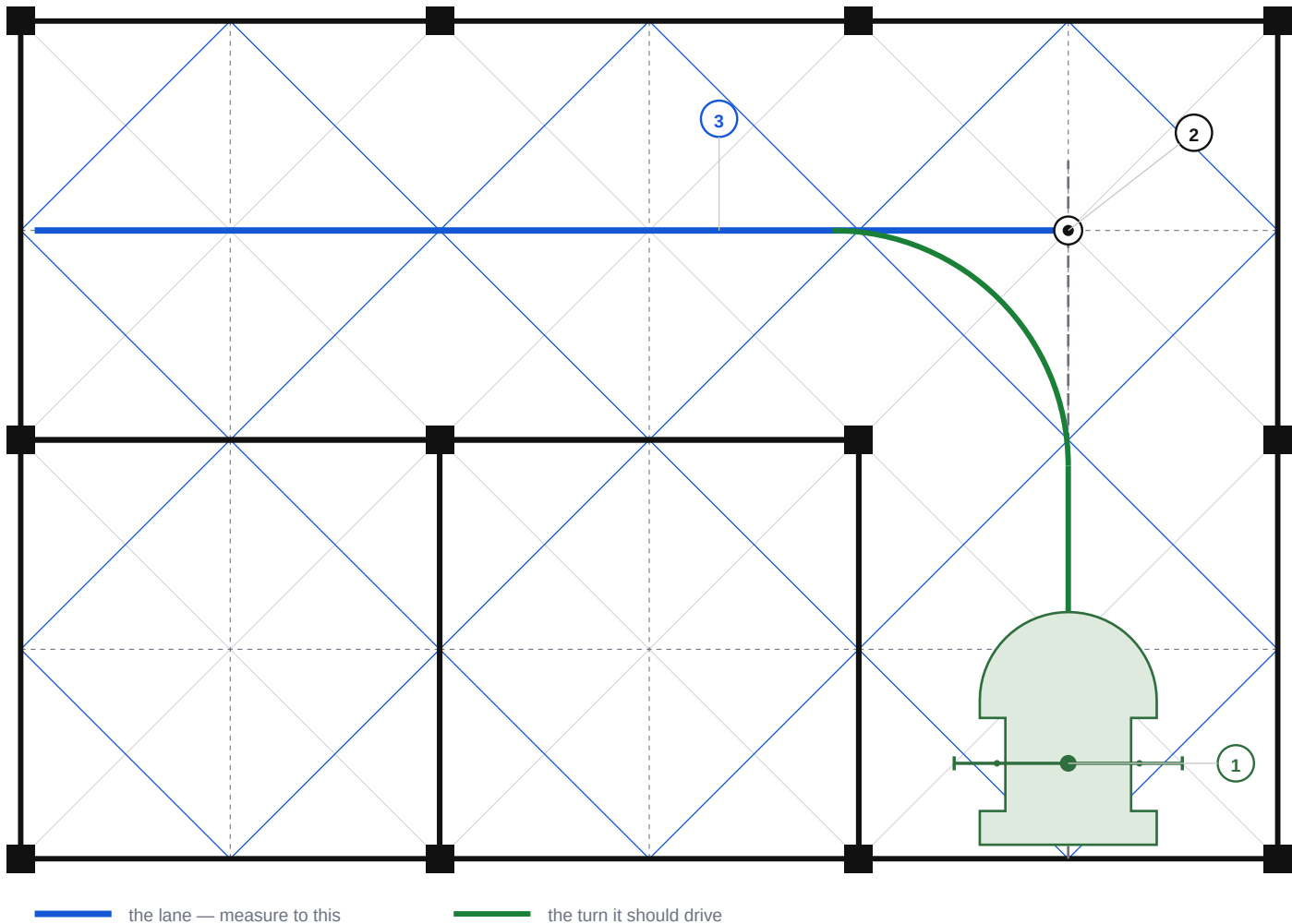
3 Coming out she should be on the blue line and square to it. Measure the wall gap either side; they should match.

This row already agrees with itself: the offset it has and the offset the radius wants are the same number, so there is no drawn error to chase. What is left is the floor.

1 mm of entry offset moves her 1.00 mm. 1 deg/s of omega moves her 0.59 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 0 — SS90EL, a 90 out of a straight

the search 90 — she stops and senses either side of it, so it need not chain



SEL,0 POS,1 ARC,300,90,170,2500,101,90 — the table holds 100; this line corrects it

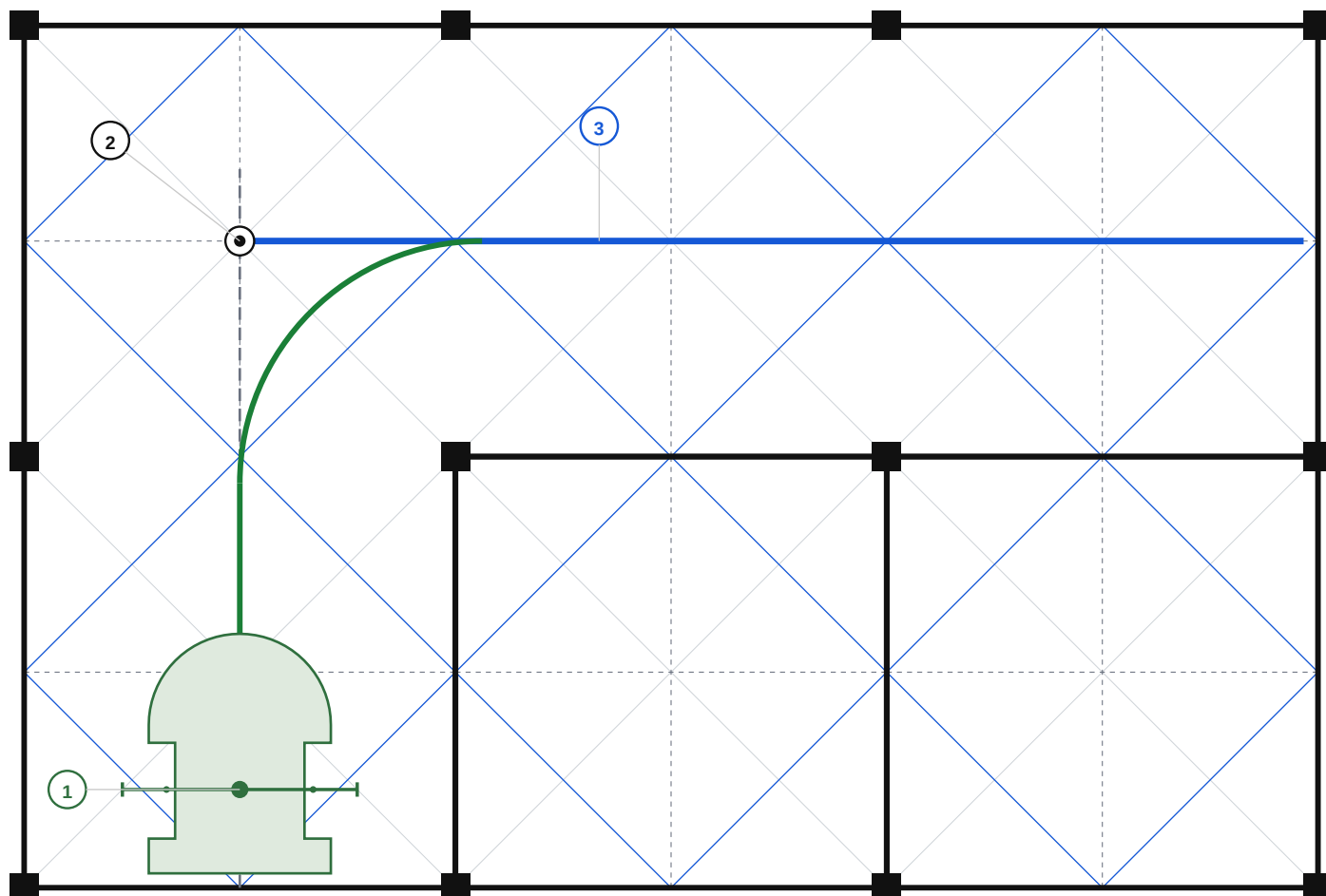
commanded speed	300 mm/s	
omega	170 deg/s	the only thing that sets the radius
$R = v / \text{omega}$	101 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(45^\circ)$	101 mm	either side of where the lanes cross
+ half cell	0 mm	two straight lanes cross AT the cell centre, so nothing is added
entry_offset it needs	101 mm	so the arc is tangent to both lanes
entry_offset it has	100 mm	out by -1 mm
lead-in it drives now	129 mm	out of the start cell, then the arc
lead-in once corrected	128 mm	she starts turning 1 mm earlier

What to read off the floor

- Back her against the wall, facing north. The marked line across her is the AXLE LINE — her centre of rotation, and the point every distance on this page is measured from.
 - The two lanes cross AT the cell centre for a 90, so there is no half cell to add here — the tangent is the whole offset.
 - Coming out she should be on the blue line and square to it. Measure the wall gap either side; they should match.
- This row already agrees with itself: the offset it has and the offset the radius wants are the same number, so there is no drawn error to chase. What is left is the floor.
- 1 mm of entry offset moves her 1.00 mm. 1 deg/s of omega moves her 0.59 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 3 — SS90R, a 90 out of a straight

the speed-run 90 — this one has to chain, wall midpoint to wall midpoint



the lane — measure to this

the turn it should drive

SEL,3 POS,1 ARC,300,-90,170,2500,101,90 — the table holds 100; this line corrects it

commanded speed	300 mm/s	
omega	170 deg/s	the only thing that sets the radius
$R = v / \omega$	101 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(45^\circ)$	101 mm	either side of where the lanes cross
+ half cell	0 mm	two straight lanes cross AT the cell centre, so nothing is added
entry_offset it needs	101 mm	so the arc is tangent to both lanes
entry_offset it has	100 mm	out by -1 mm
lead-in it drives now	129 mm	out of the start cell, then the arc
lead-in once corrected	128 mm	she starts turning 1 mm earlier

What to read off the floor

1 Back her against the wall, facing north. The marked line across her is the AXLE LINE — her centre of rotation, and the point every distance on this page is measured from.

2 The two lanes cross AT the cell centre for a 90, so there is no half cell to add here — the tangent is the whole offset.

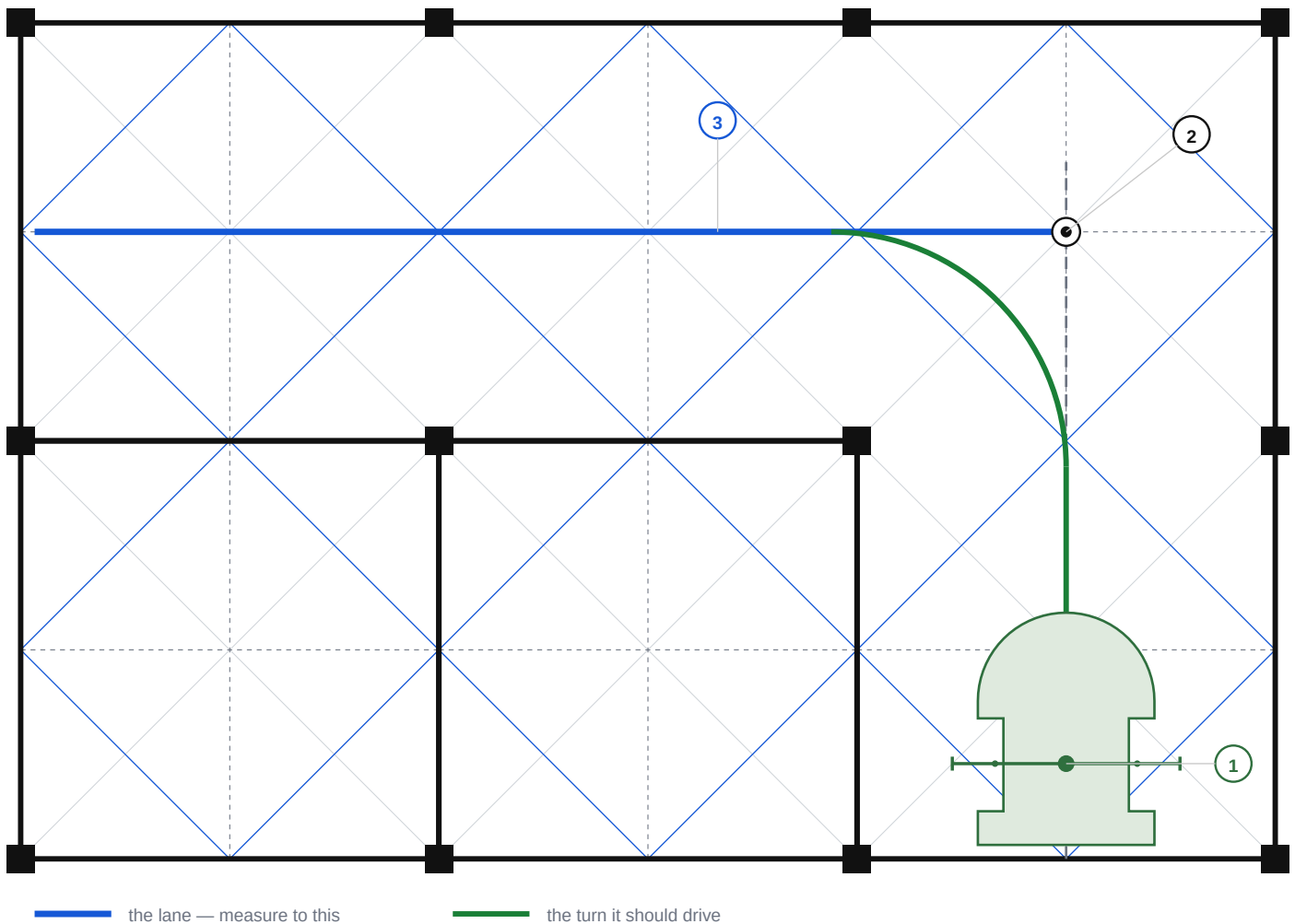
3 Coming out she should be on the blue line and square to it. Measure the wall gap either side; they should match.

This row already agrees with itself: the offset it has and the offset the radius wants are the same number, so there is no drawn error to chase. What is left is the floor.

1 mm of entry offset moves her 1.00 mm. 1 deg/s of omega moves her 0.59 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 2 — SS90L, a 90 out of a straight

the speed-run 90 — this one has to chain, wall midpoint to wall midpoint



SEL,2 POS,1 ARC,300,90,170,2500,101,90 — the table holds 100; this line corrects it

commanded speed	300 mm/s	
omega	170 deg/s	the only thing that sets the radius
$R = v / \omega$	101 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(45^\circ)$	101 mm	either side of where the lanes cross
+ half cell	0 mm	two straight lanes cross AT the cell centre, so nothing is added
entry_offset it needs	101 mm	so the arc is tangent to both lanes
entry_offset it has	100 mm	out by -1 mm
lead-in it drives now	129 mm	out of the start cell, then the arc
lead-in once corrected	128 mm	she starts turning 1 mm earlier

What to read off the floor

1 Back her against the wall, facing north. The marked line across her is the AXLE LINE — her centre of rotation, and the point every distance on this page is measured from.

2 The two lanes cross AT the cell centre for a 90, so there is no half cell to add here — the tangent is the whole offset.

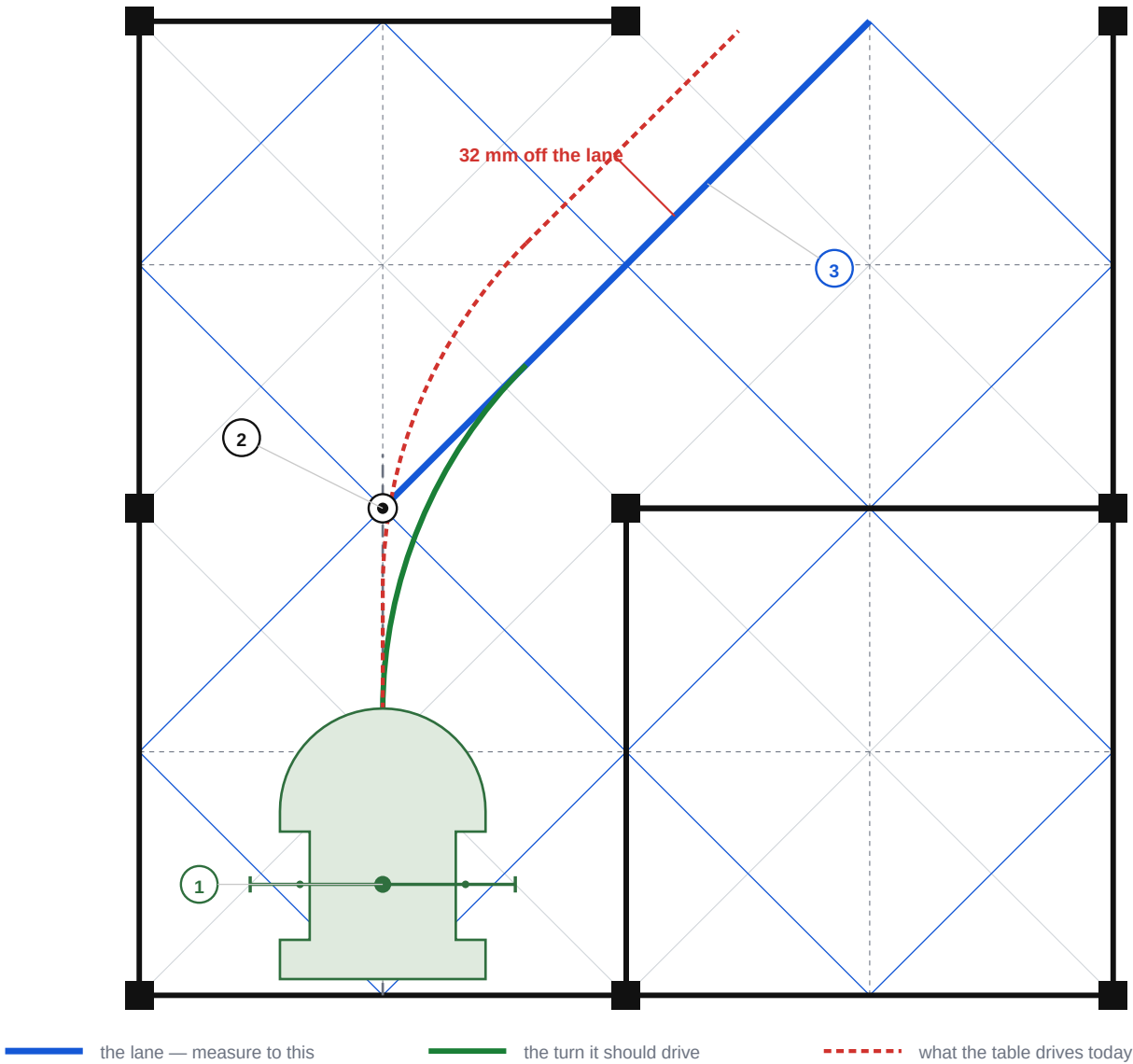
3 Coming out she should be on the blue line and square to it. Measure the wall gap either side; they should match.

This row already agrees with itself: the offset it has and the offset the radius wants are the same number, so there is no drawn error to chase. What is left is the floor.

1 mm of entry offset moves her 1.00 mm. 1 deg/s of omega moves her 0.59 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 7 — SD45R, straight onto the diagonal

one turn on its own — the heavy blue line is the thing she has to end up on



SEL,7 POS,1 ARC,300,-45,95,2500,165,90 — the table holds 120; this line corrects it

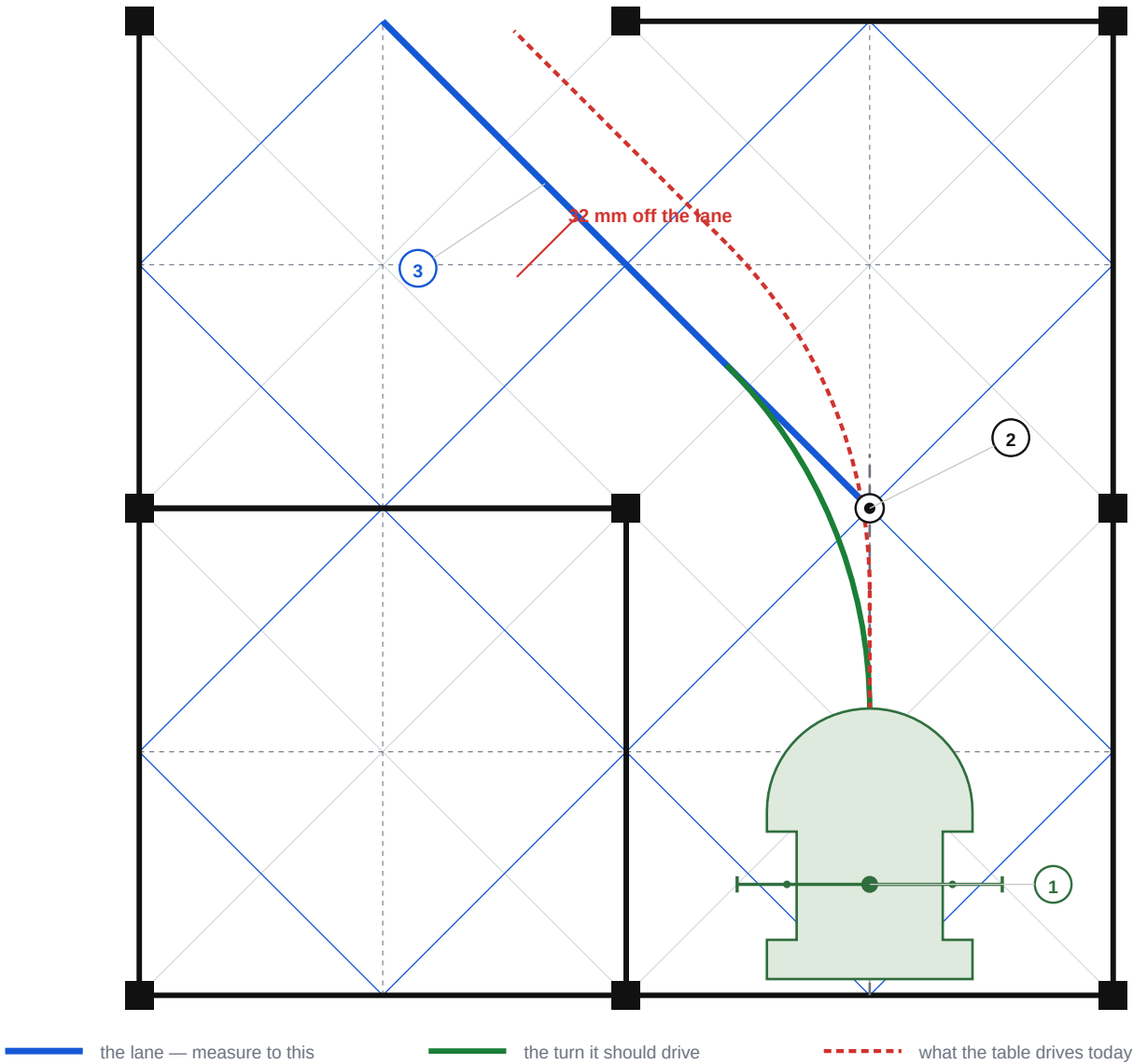
commanded speed	300 mm/s	
omega	95 deg/s	the only thing that sets the radius
$R = v / \omega$	181 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(22.5^\circ)$	75 mm	either side of where the lanes cross
+ half cell	90 mm	a straight lane meets a diagonal at the WALL MIDPOINT, not the cell centre
entry_offset it needs	165 mm	so the arc is tangent to both lanes
entry_offset it has	120 mm	out by -45 mm
lead-in it drives now	109 mm	out of the start cell, then the arc
lead-in once corrected	64 mm	she starts turning 45 mm earlier

What to read off the floor

- Back her against the wall, facing north. The marked line across her is the AXLE LINE — her centre of rotation, and the point every distance on this page is measured from. 35 mm of chassis sits behind it.
 - She should already be turning by the time she reaches the wall midpoint — that, not the cell centre, is where the lanes cross.
 - Coming out of the arc she should be ON the blue line and parallel to it. That is the whole measurement; ignore where she stops. Too LATE (offset too small) and she overshoots to the far side of the line, away from the post; too EARLY and she cuts inside it, close to that post. She is LATE today, by 45 mm of offset.
- 1 mm of entry offset moves her 0.71 mm. 1 deg/s of omega moves her 0.56 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 6 — SD45L, straight onto the diagonal

one turn on its own — the heavy blue line is the thing she has to end up on



SEL,6 POS,1 ARC,300,45,95,2500,165,90 — the table holds 120; this line corrects it

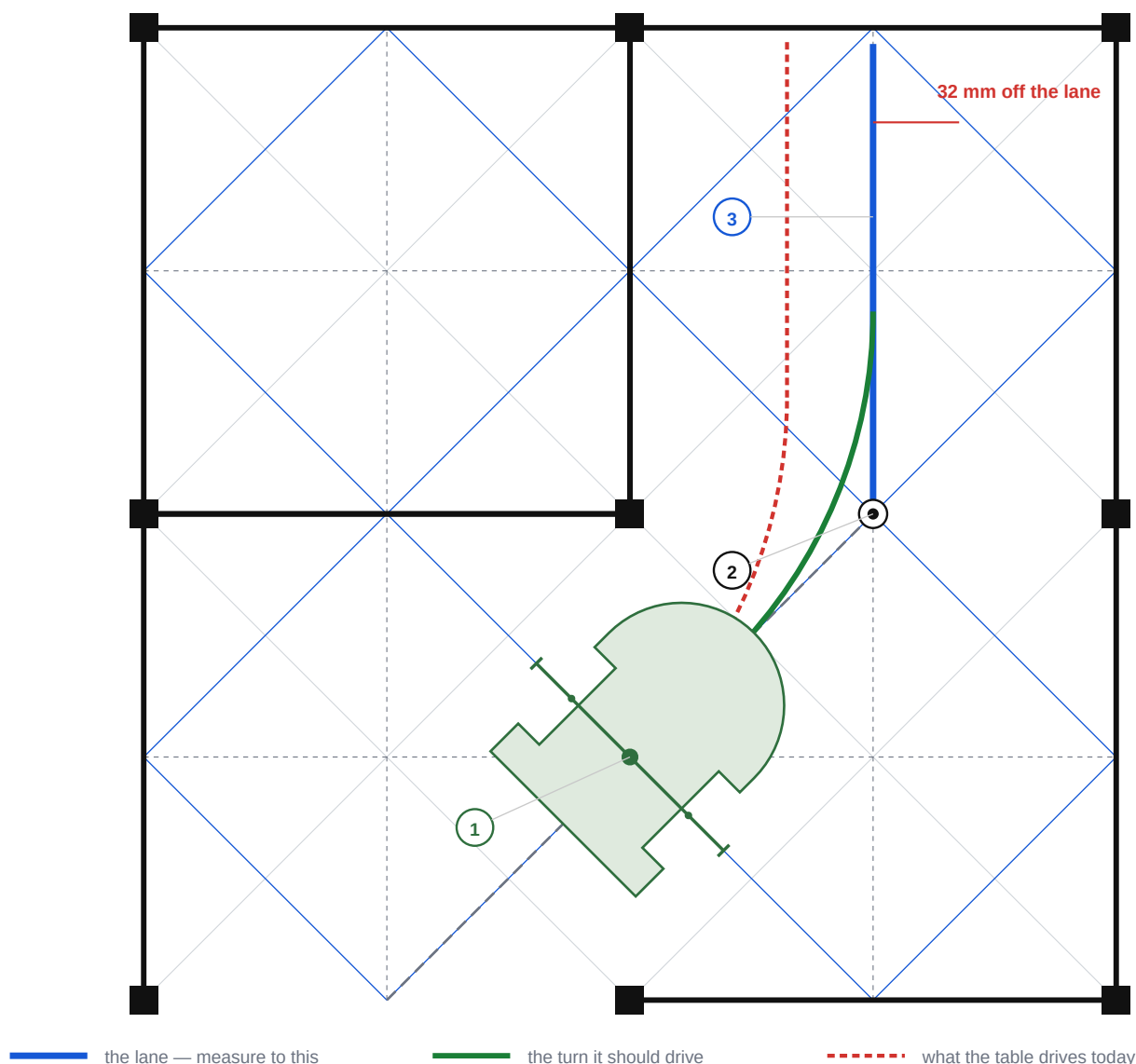
commanded speed	300 mm/s	
omega	95 deg/s	the only thing that sets the radius
$R = v / \omega$	181 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(22.5^\circ)$	75 mm	either side of where the lanes cross
+ half cell	90 mm	a straight lane meets a diagonal at the WALL MIDPOINT, not the cell centre
entry_offset it needs	165 mm	so the arc is tangent to both lanes
entry_offset it has	120 mm	out by -45 mm
lead-in it drives now	109 mm	out of the start cell, then the arc
lead-in once corrected	64 mm	she starts turning 45 mm earlier

What to read off the floor

- Back her against the wall, facing north. The marked line across her is the AXLE LINE — her centre of rotation, and the point every distance on this page is measured from. 35 mm of chassis sits behind it.
 - She should already be turning by the time she reaches the wall midpoint — that, not the cell centre, is where the lanes cross.
 - Coming out of the arc she should be ON the blue line and parallel to it. That is the whole measurement; ignore where she stops. Too LATE (offset too small) and she overshoots to the far side of the line, away from the post; too EARLY and she cuts inside it, close to that post. She is LATE today, by 45 mm of offset.
- 1 mm of entry offset moves her 0.71 mm. 1 deg/s of omega moves her 0.56 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 8 — DS45L, diagonal back onto a straight

she is placed ON the diagonal for this one — there is no wall to back up against



SEL,8 POS,1 ARC,300,45,95,2500,75,90 — the table holds 120; this line corrects it

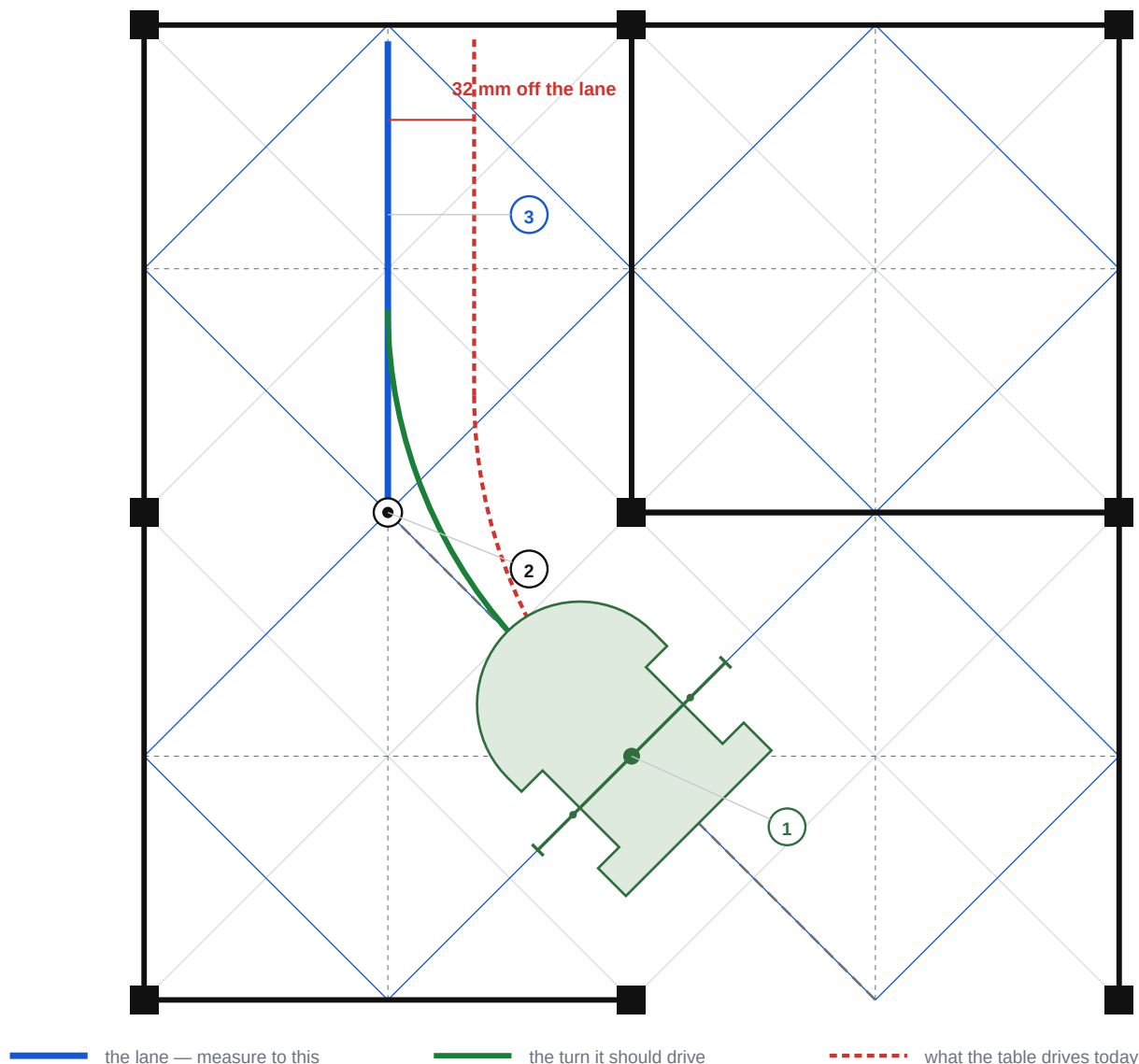
commanded speed	300 mm/s	
omega	95 deg/s	the only thing that sets the radius
$R = v / \omega$	181 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(22.5^\circ)$	75 mm	either side of where the lanes cross
+ half cell	0 mm	she turns AT her lattice point here, so nothing is added
entry_offset it needs	75 mm	so the arc is tangent to both lanes
entry_offset it has	120 mm	out by +45 mm
lead-in it drives now	7 mm	along the diagonal, then the arc
lead-in once corrected	52 mm	she starts turning 45 mm later

What to read off the floor

- Put her down so the AXLE LINE — the line through the motors, which is her centre of rotation — crosses the blue diagonal at the marked point, square to it. Her body will straddle the gap; that is right.
- The diagonal and the straight lane cross AT the wall midpoint, which is also the point the offset is measured from.
- Coming out she should be on the centreline of the column above, and square to it.
The diagonal runs 63.6 mm from each post centre, so 55 mm clear of the nearest post CORNER either side — a 110 mm channel for a 61.2 mm track. Square her carefully: a degree of yaw here is worth more than a millimetre of position.
Too LATE (offset too small) and she overshoots to the far side of the line, away from the post; too EARLY and she cuts inside it, close to that post. She is EARLY today, by 45 mm of offset.
1 mm of entry offset moves her 0.71 mm. 1 deg/s of omega moves her 0.56 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 9 — DS45R, diagonal back onto a straight

she is placed ON the diagonal for this one — there is no wall to back up against



SEL,9 POS,1 ARC,300,-45,95,2500,75,90 — the table holds 120; this line corrects it

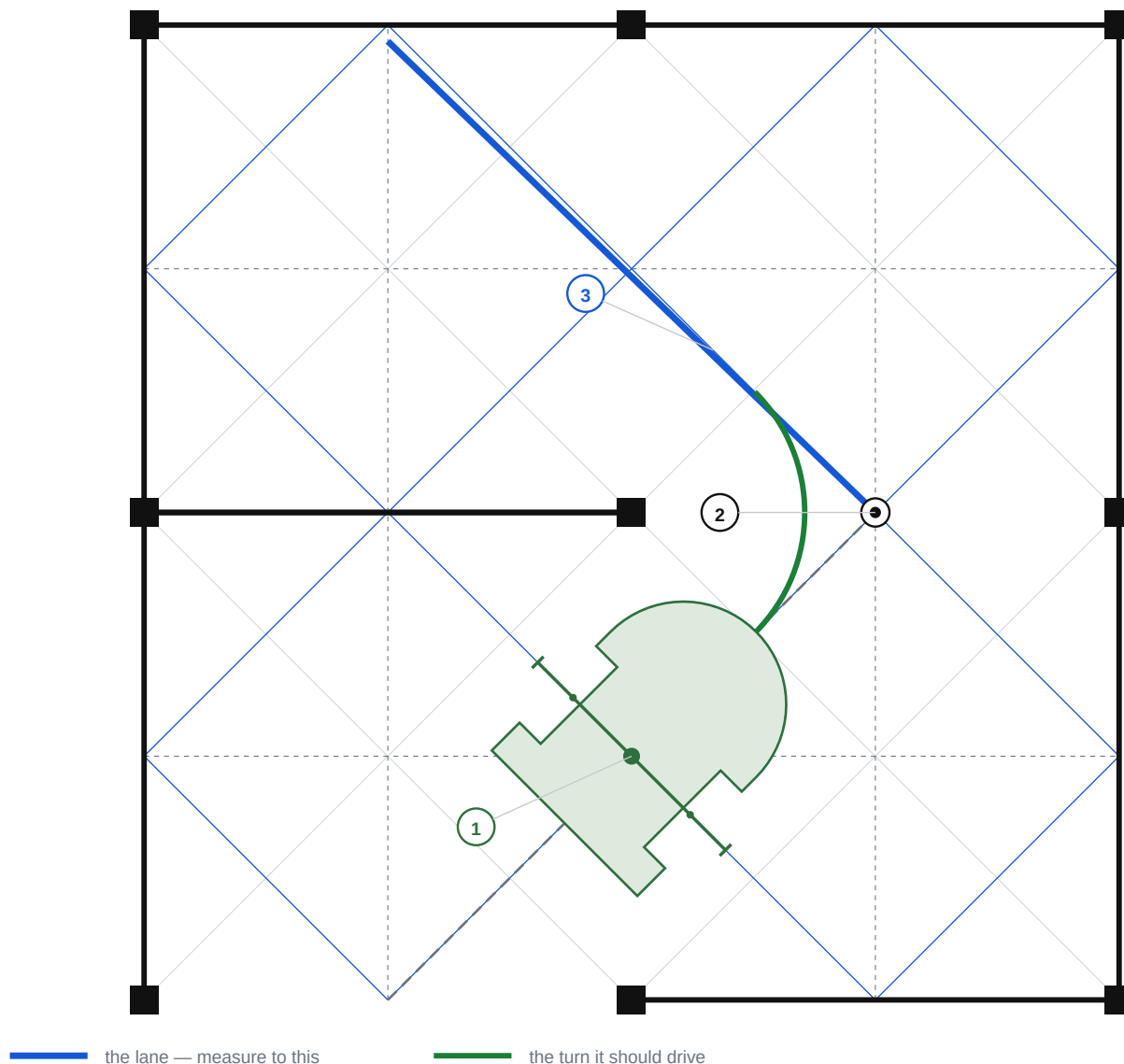
commanded speed	300 mm/s	
omega	95 deg/s	the only thing that sets the radius
$R = v / \omega$	181 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(22.5^\circ)$	75 mm	either side of where the lanes cross
+ half cell	0 mm	she turns AT her lattice point here, so nothing is added
entry_offset it needs	75 mm	so the arc is tangent to both lanes
entry_offset it has	120 mm	out by +45 mm
lead-in it drives now	7 mm	along the diagonal, then the arc
lead-in once corrected	52 mm	she starts turning 45 mm later

What to read off the floor

- Put her down so the AXLE LINE — the line through the motors, which is her centre of rotation — crosses the blue diagonal at the marked point, square to it. Her body will straddle the gap; that is right.
- The diagonal and the straight lane cross AT the wall midpoint, which is also the point the offset is measured from.
- Coming out she should be on the centreline of the column above, and square to it.
The diagonal runs 63.6 mm from each post centre, so 55 mm clear of the nearest post CORNER either side — a 110 mm channel for a 61.2 mm track. Square her carefully: a degree of yaw here is worth more than a millimetre of position.
Too LATE (offset too small) and she overshoots to the far side of the line, away from the post; too EARLY and she cuts inside it, close to that post. She is EARLY today, by 45 mm of offset.
1 mm of entry offset moves her 0.71 mm. 1 deg/s of omega moves her 0.56 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 14 — DD90L, the diagonal changes hand

the only turn that begins and ends on a diagonal — and the only one with 127 mm, not 180, either side of it



SEL,14 POS,1 ARC,300,90,273,2500,63,90

commanded speed	300 mm/s	
omega	273 deg/s	the only thing that sets the radius
$R = v / \omega$	63 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(45^\circ)$	63 mm	either side of where the lanes cross
+ half cell	0 mm	two diagonals cross AT the wall midpoint, so nothing is added
entry_offset it needs	63 mm	so the arc is tangent to both lanes
entry_offset it has	63 mm	agrees
lead-in it drives now	64 mm	along the diagonal, then the arc

What to read off the floor

1 Put her down so the AXLE LINE — the line through the motors, which is her centre of rotation — crosses the blue diagonal at the marked point, square to it. Her body will straddle the gap; that is right.

2 She turns at the wall midpoint, and the step either side is 127.279 mm — not 180. That is what makes this the tight one.

3 Coming out she should be on the OTHER blue diagonal and parallel to it, having passed the post without touching it.

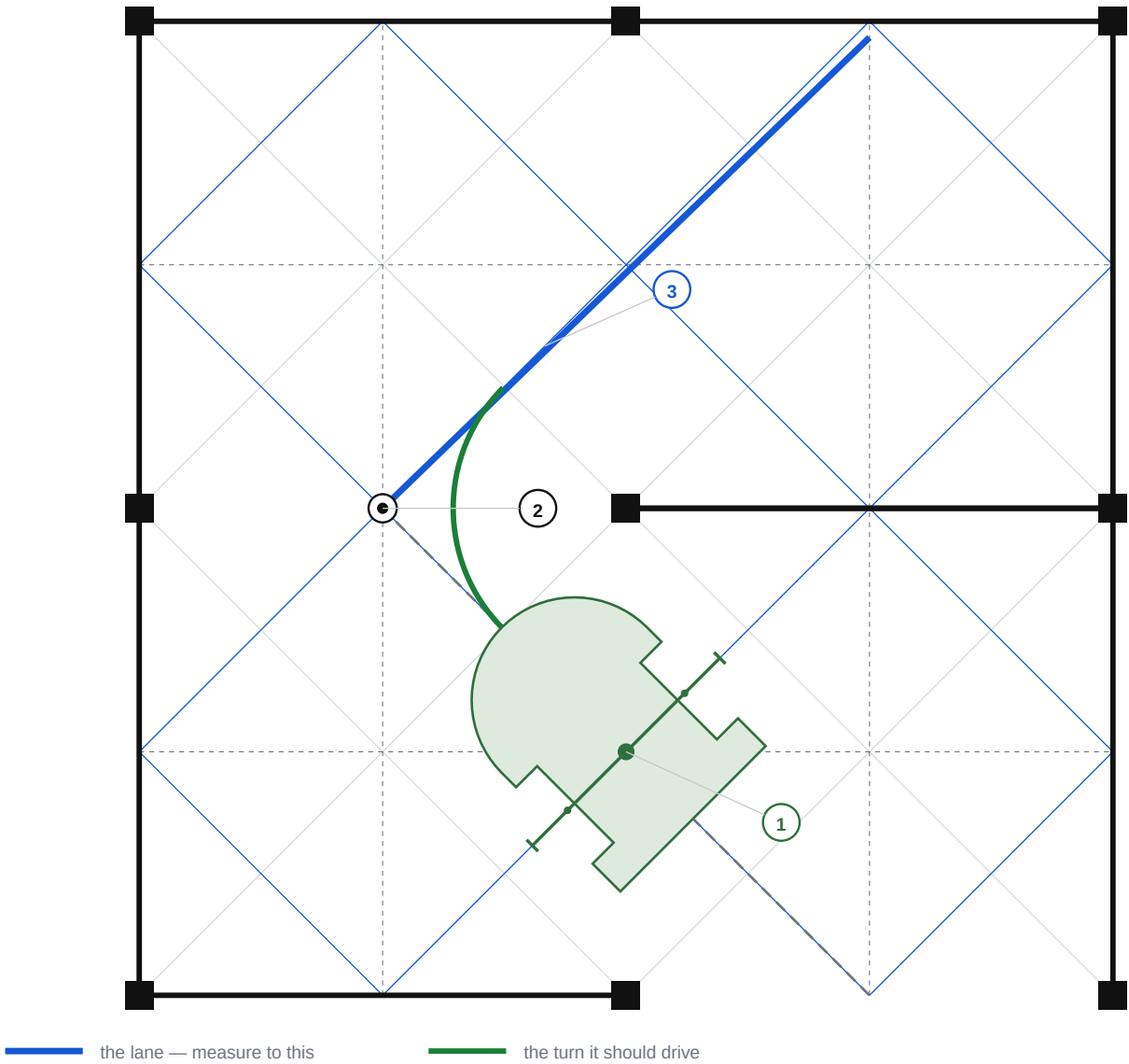
The diagonal runs 63.6 mm from each post centre, so 55 mm clear of the nearest post CORNER either side — a 110 mm channel for a 61.2 mm track. Square her carefully: a degree of yaw here is worth more than a millimetre of position.

This row already agrees with itself: the offset it has and the offset the radius wants are the same number, so there is no drawn error to chase. What is left is the floor.

1 mm of entry offset moves her 1.00 mm. 1 deg/s of omega moves her 0.23 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 15 — DD90R, the diagonal changes hand

the only turn that begins and ends on a diagonal — and the only one with 127 mm, not 180, either side of it



SEL,15 POS,1 ARC,300,-90,273,2500,63,90

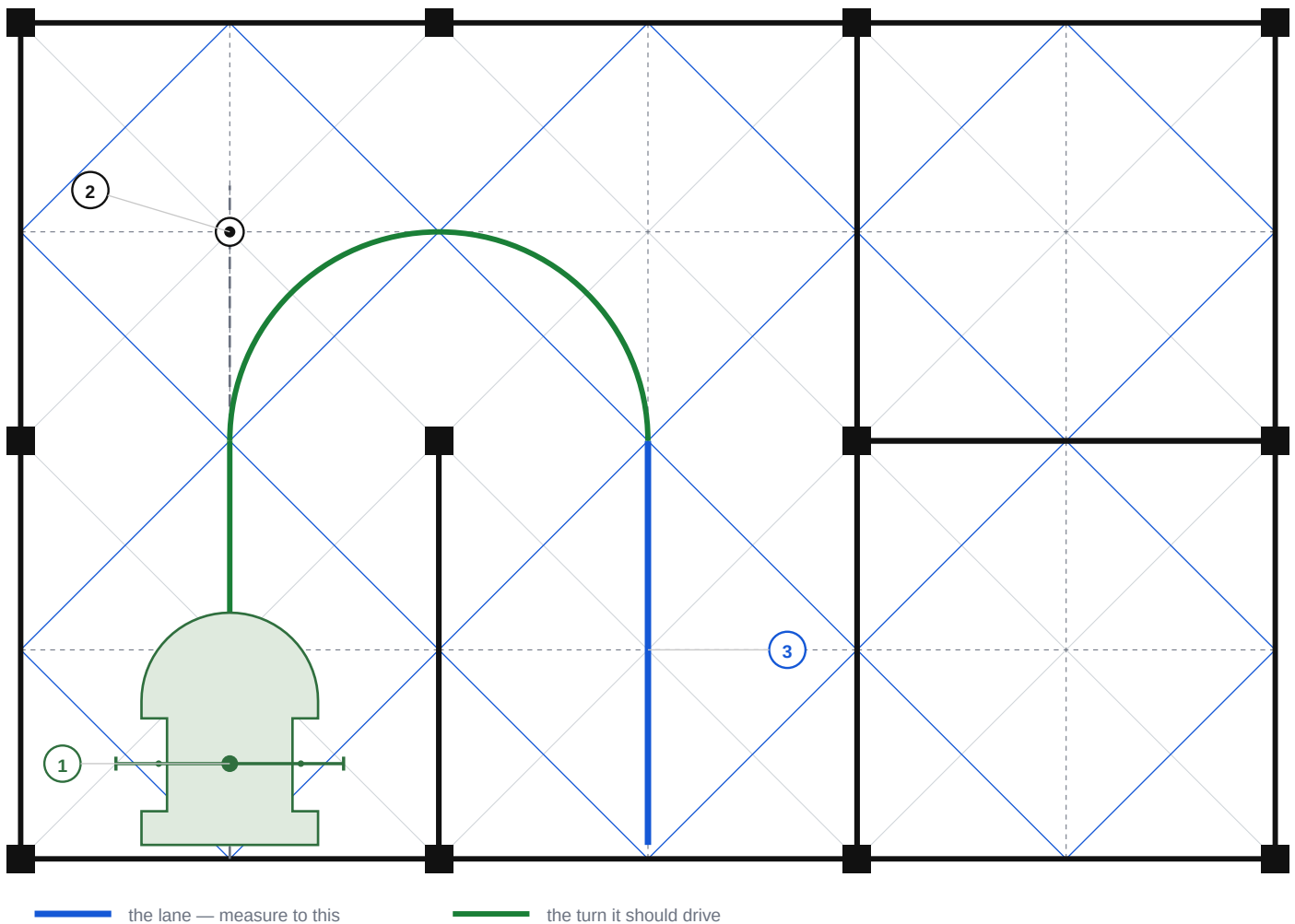
commanded speed	300 mm/s	
omega	273 deg/s	the only thing that sets the radius
$R = v / \omega$	63 mm	instantaneous; alpha's ramps make the real arc a little wider
tangent $R \cdot \tan(45^\circ)$	63 mm	either side of where the lanes cross
+ half cell	0 mm	two diagonals cross AT the wall midpoint, so nothing is added
entry_offset it needs	63 mm	so the arc is tangent to both lanes
entry_offset it has	63 mm	agrees
lead-in it drives now	64 mm	along the diagonal, then the arc

What to read off the floor

- Put her down so the AXLE LINE — the line through the motors, which is her centre of rotation — crosses the blue diagonal at the marked point, square to it. Her body will straddle the gap; that is right.
 - She turns at the wall midpoint, and the step either side is 127.279 mm — not 180. That is what makes this the tight one.
 - Coming out she should be on the OTHER blue diagonal and parallel to it, having passed the post without touching it.
- The diagonal runs 63.6 mm from each post centre, so 55 mm clear of the nearest post CORNER either side — a 110 mm channel for a 61.2 mm track. Square her carefully: a degree of yaw here is worth more than a millimetre of position.
- This row already agrees with itself: the offset it has and the offset the radius wants are the same number, so there is no drawn error to chase. What is left is the floor.
- 1 mm of entry offset moves her 1.00 mm. 1 deg/s of omega moves her 0.23 mm. Trim the OFFSET: omega sets the radius, and the radius has to fit the space.

Row 5 — SS180R, the hairpin

not a dead end — she comes back down the NEXT lane, so the lateral step has to be exactly one cell



SEL,5 POS,1 ARC,300,-180,191,2500,90,90

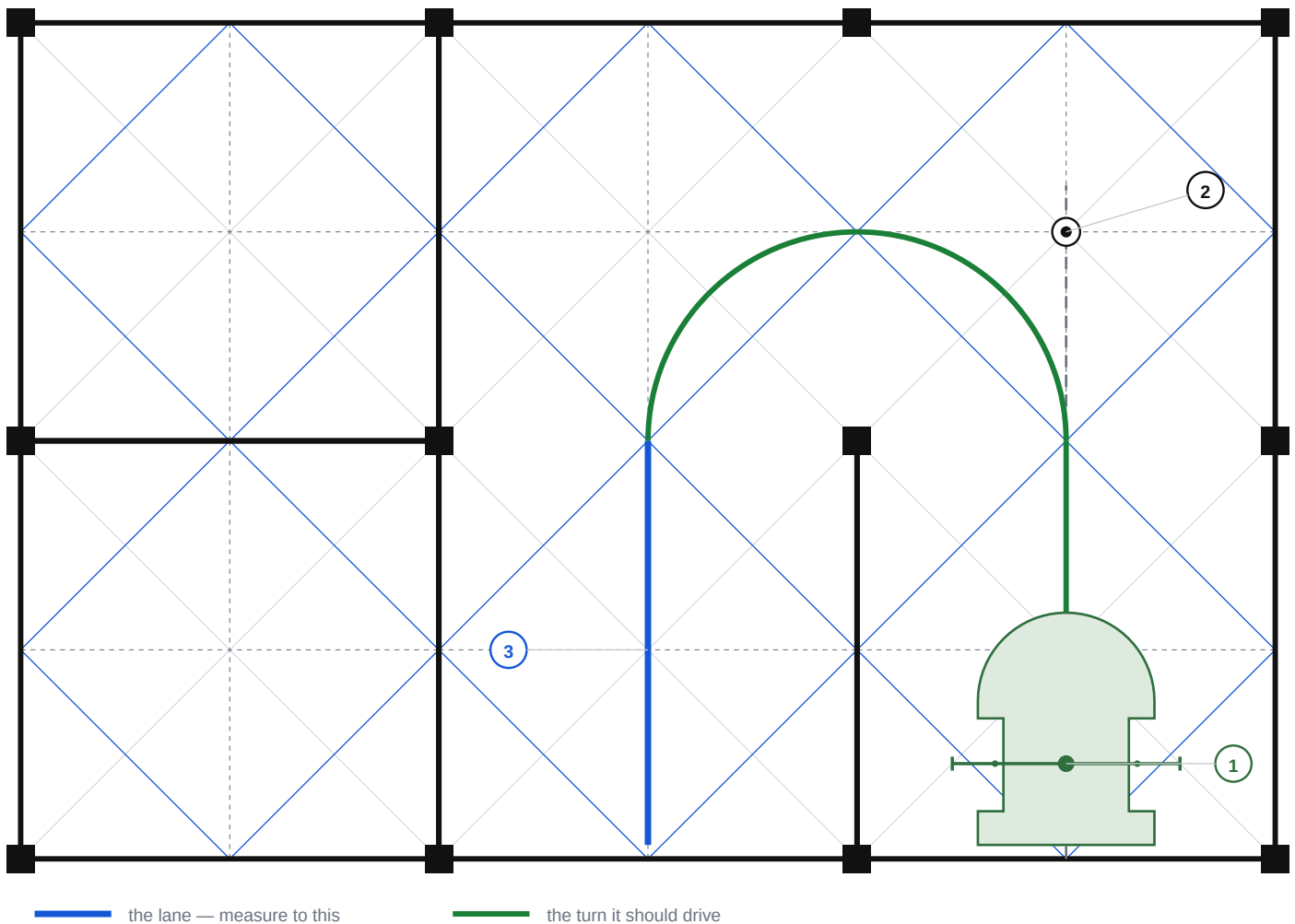
commanded speed	300 mm/s	
omega	191 deg/s	and this one is NOT free — see below
$R = v / \omega$	90 mm	instantaneous; alpha's ramps make the real arc wider
lateral step 2R	180 mm	has to be 180 — one cell — or she lands off the lane
out by	0 mm	— it lands on the lane
entry_offset	90 mm	sets only how far up the cell she does it
lead-in it drives	139 mm	out of the start cell, then the arc

What to read off the floor

- 1 Back her against the wall, facing north. The line across her is the AXLE LINE, her centre of rotation and the datum for everything here. 2 entry_offset is measured from the axle line to the start of the arc.
 - 3 She must come back down the blue line — the next lane over, 180 mm across. Measure the wall gap on both sides.
- OMEGA IS NOT FREE ON THIS ROW. A 180 of radius R steps her sideways by 2R, and that has to be exactly one cell.
 $2R = 180$ forces $R = 90$ and $\omega = 191$ deg/s at 300 mm/s. 1 deg/s moves her 0.94 mm; the offset moves her not at all.

Row 4 — SS180L, the hairpin

not a dead end — she comes back down the NEXT lane, so the lateral step has to be exactly one cell



SEL,4 POS,1 ARC,300,180,191,2500,90,90

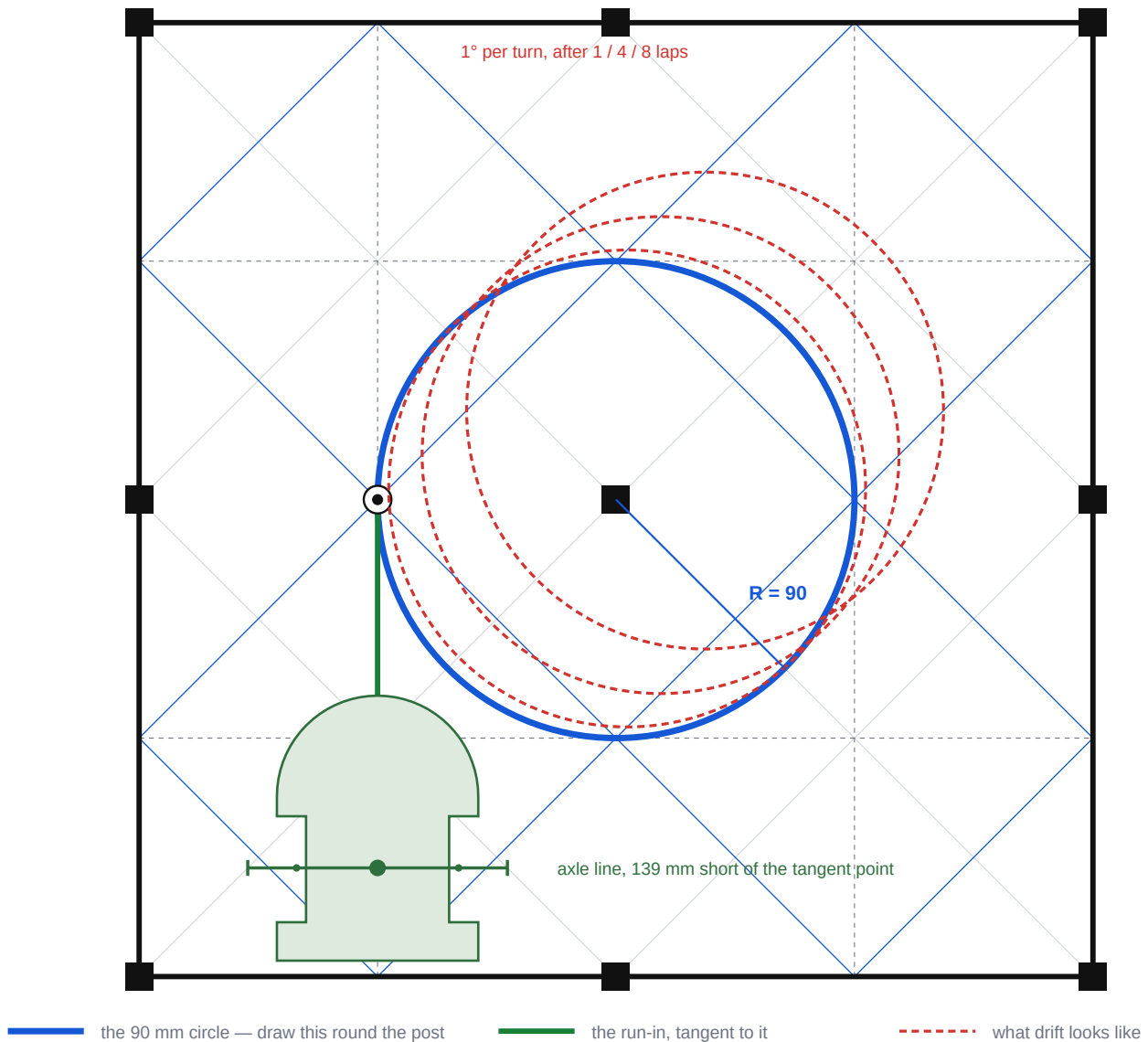
commanded speed	300 mm/s	
omega	191 deg/s	and this one is NOT free — see below
$R = v / \omega$	90 mm	instantaneous; alpha's ramps make the real arc wider
lateral step $2R$	180 mm	has to be 180 — one cell — or she lands off the lane
out by	0 mm	— it lands on the lane
entry_offset	90 mm	sets only how far up the cell she does it
lead-in it drives	139 mm	out of the start cell, then the arc

What to read off the floor

- 1 Back her against the wall, facing north. The line across her is the AXLE LINE, her centre of rotation and the datum for everything here. 2 entry_offset is measured from the axle line to the lane.
- 3 She must come back down the blue line — the next lane over, 180 mm across. Measure the wall gap on both sides. OMEGA IS NOT FREE ON THIS ROW. A 180 of radius R steps her sideways by $2R$, and that has to be exactly one cell. $2R = 180$ forces $R = 90$ and $\omega = 191$ deg/s at 300 mm/s. 1 deg/s moves her 0.94 mm; the offset moves her not at all.

The post loop — four chained 90s, driven as one circle

the rig that pins omega for a chainable 90, and shows heading error a tenth of a degree at a time



LOOP,8,191,2500,1,300

then run Post loop (Calibration, or BT key A)

commanded speed	300 mm/s	
omega	191 deg/s	NOT free: $R = v/\omega$ and R has to be 90
$R = v / \omega$	90 mm	half a cell — the quarter circle that chains
one lap	565 mm	$2\pi R$. Odometry over N laps gives R back with no ruler.
lead-in from the wall	139 mm	$49 + 90$: out of the start cell to the tangent point
at the SS90 row's 170	635 mm	R 101: a lap 70 mm long, and 22 mm of overlap between consecutive turns
and with the ramps in	~700 mm	the zigzag figure — which is why driving it continuously is the cleaner test
drift per lap	6 mm	for one degree of heading error per turn

What to read off the floor

Four chained same-hand 90s of radius 90 ARE a circle centred on a post: the arc centre sits 90 mm to her right, which is the post, and it is the same point every quarter. So this rig and "four chained turns" are the same experiment, and because she drives it continuously there are no ramps between the quarters — the ideal $R = v/\omega$ applies and omega is pinned at 191.

OVER A CLOSED LAP EVERYTHING EXCEPT HEADING CANCELS. The four local frames sit at 0, 90, 180 and 270 degrees, so a consistent radius error, offset error or tangential lag sums to zero. A heading error does not: the frames precess, the lap stops closing, and the centre walks about 6 mm a lap per degree per turn. Eight laps turn a tenth of a degree into five millimetres.

So: omega sets the SIZE of the circle — read it off the drawn one, or off odometry. Alpha and the gyro scale set the DRIFT of its centre. Tune omega until she tracks the circle, then alpha until the centre stops walking. The gyro will report the commanded angle either way, because the loop closes on it; the floor is the only witness.

Rows 10–13 — the 135s

why there is no page for them yet

A 135 turns a straight onto a diagonal the long way round, or back off it. The lanes still cross at a wall midpoint, so SD135 carries the same half cell as SD45 does. What it does not have is room.

The tangent of a 135 turn is $R \cdot \tan(67.5^\circ) = 2.414R$ — nearly six times an SD45's. At the table's omega of 191 deg/s, R is 90 mm and the tangent is 217 mm. Add the half cell and an SD135 would have to start turning 307 mm before the cell it is booked in at, which is most of two cells back, through a wall.

To make the tangent fit inside one cell the radius has to come down to about 37 mm, which at 300 mm/s is omega 464 deg/s. That is a spin with the wheels rolling, and the lateral acceleration goes with it.

So the 135s stay as slots. The planner does not emit them, nothing drives them, and the four rows in the table are arithmetic holding a place. When they are wanted, they want their own speed as well as their own radius — which is a different page from any of these.